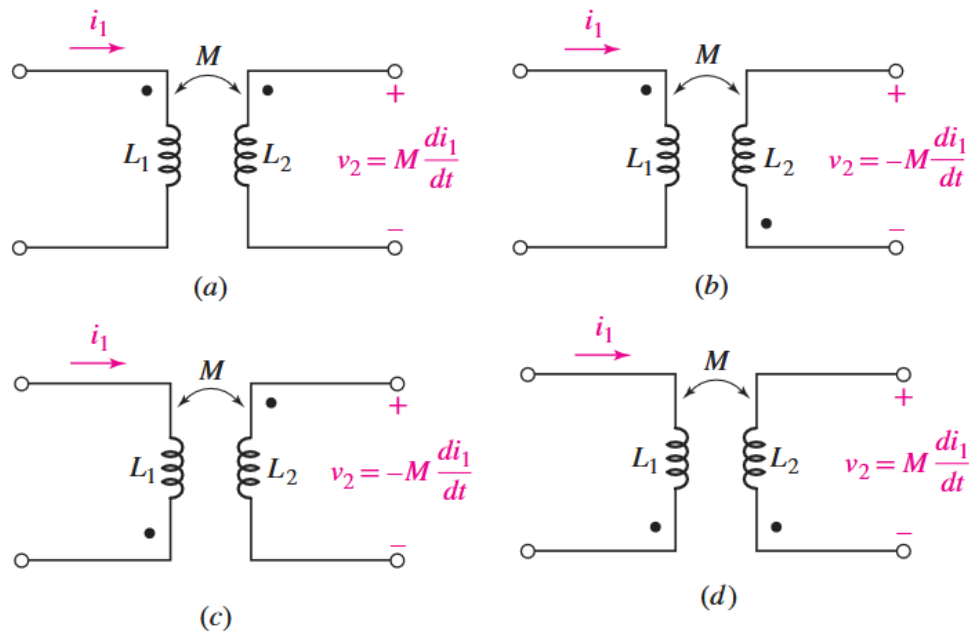


HW 9 - Zeshui Song

Notes:

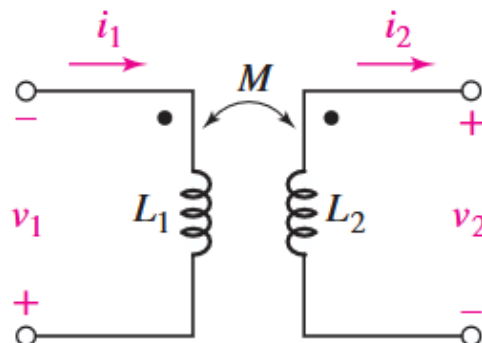
- **Dot Convention for Mutual Inductance:**

- A current entering the *dotted* terminal of one coil produces an open-circuit voltage with a positive voltage reference at the *dotted* terminal of the second coil.
- A current entering the *undotted* terminal of one coil provides a voltage that is positively sensed at the *undotted* terminal of the second coil



Note that **self-induction** exists if the current in the right loop (i_2) is non-zero and would cause back emf.

- Combining mutual and self-induction:



By passive current convention, current i_1 should enter the positive terminal of the inductor L_1 . However, the polarity of v_1 indicates that di_1/dt must be negative (indicating current flowing in the opposite direction). By the dot convention, current i_2 enters the undotted terminal of L_2 , so the mutual inductance contribution to v_1 is positive by the polarity of v_1 .

$$v_1 = -L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

Similarly for v_2 :

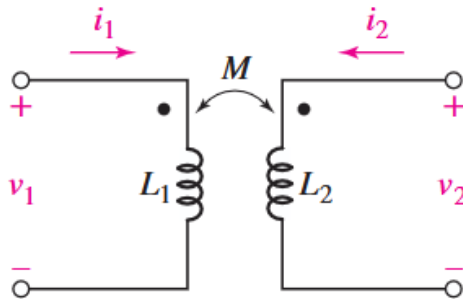
$$v_2 = -L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

These same sign conventions apply to sinusoidal steady-state analysis, where i_1 and i_2 are phasors.

$$\mathbf{V}_1 = -j\omega L_1 \mathbf{I}_1 + j\omega M \mathbf{I}_2$$

$$\mathbf{V}_2 = -j\omega L_2 \mathbf{I}_2 + j\omega M \mathbf{I}_1$$

- Total energy in coupled inductors:



$$w(t) = \frac{1}{2} L_1 [i_1(t)]^2 + \frac{1}{2} L_2 [i_2(t)]^2 + M [i_1(t)][i_2(t)]$$

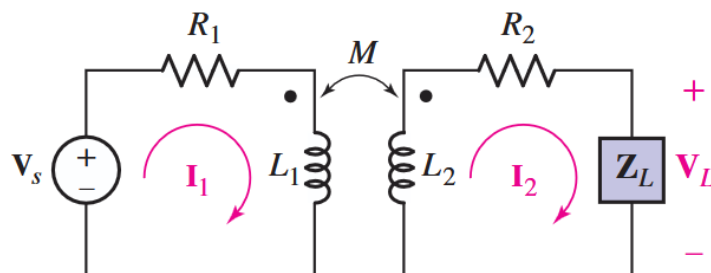
Limit on M:

$$M \leq \sqrt{L_1 L_2}$$

Coupling coefficient:

$$k = \frac{M}{\sqrt{L_1 L_2}} \implies 0 \leq k \leq 1$$

- Linear transformer and reflected impedance:



The *primary circuit* contains the source v_s and the *secondary circuit* contains the load $\mathbf{V}_L$.

By using mesh analysis on each of the two loops, we can simplify the circuit to an equivalent circuit with a $\mathbf{V}_s$ in series with a reflected impedance $\mathbf{Z}_{\text{in}}$:

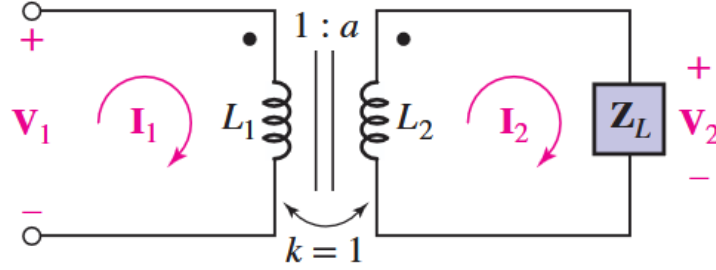
$$\frac{\mathbf{V}_s}{\mathbf{I}_1} = \mathbf{Z}_{\text{in}} = \mathbf{Z}_{11} - \frac{(j\omega)^2 M^2}{\mathbf{Z}_{22}}$$

Where $\mathbf{Z}_{11}$ and $\mathbf{Z}_{22}$ are the self-impedances of the primary and secondary circuits, respectively.

$$\mathbf{Z}_{11} = R_1 + j\omega L_1$$

$$\mathbf{Z}_{22} = R_2 + j\omega L_2 + \mathbf{Z}_L$$

- **Ideal Transformer:**



In an ideal transformer, the coupling coefficient $k = 1$ and the mutual inductance $M = \sqrt{L_1 L_2}$.

Turns Ratio:

$$a^2 = \frac{L_2}{L_1} = \frac{N_2^2}{N_1^2}$$

Using turns ratio, we can have the following relationships:

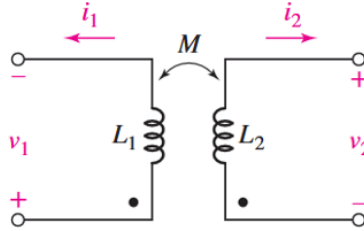
$$\frac{1}{a} = \frac{I_2}{I_1}$$

$$a = \frac{V_2}{V_1}$$

$$a^2 = \frac{\mathbf{Z}_L}{\mathbf{Z}_{\text{in}}}$$

13.6

6. The circuit in Fig. 13.38 has a sinusoidal input at $\omega = 1,000$ rad/s with $\mathbf{I}_1 = 3\angle 45^\circ$ A and a $100\ \Omega$ resistor attached across the terminals labeled v_2 . For the case where $L_1 = 50$ mH, $L_2 = 250$ mH, and $M = 0.75\sqrt{L_1 L_2}$, determine $\mathbf{V}_1$, $\mathbf{I}_2$, and $\mathbf{V}_2$ in phasor form.

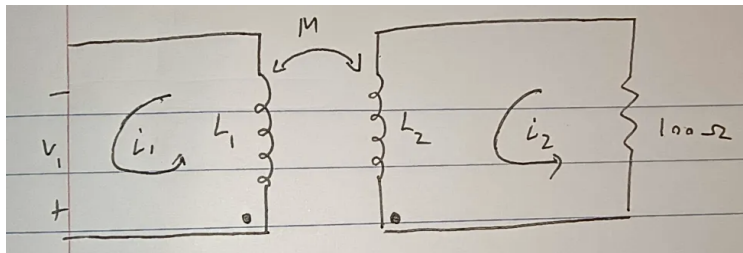


■ FIGURE 13.38

This is a circuit at sinusoidal steady state, so we can use phasor analysis. First, defining constants and impedances:

- $\omega = 1000$ rad/s
- $\mathbf{I}_1 = 3\angle 45^\circ$ A
- Resistor across v_2 : $R = 100\ \Omega$
- $L_1 = 50$ mH $\Rightarrow \mathbf{Z}_{L1} = j\omega L_1 = 50j\ \Omega$
- $L_2 = 250$ mH $\Rightarrow \mathbf{Z}_{L2} = j\omega L_2 = 250j\ \Omega$
- $M = 0.75\sqrt{L_1 L_2} = 83.852$ mH $\Rightarrow \mathbf{Z}_M = j\omega M = 83.852j\ \Omega$

Instead of following the current direction defined in the diagram, I will redraw the circuit following passive sign convention. At the end, I will map my results back to the original diagram with a sign change if necessary.



KVL to mesh 1:

$$\mathbf{V}_1 - \mathbf{Z}_{L1}\mathbf{I}_1 + \mathbf{Z}_M\mathbf{I}_2 = 0$$

$$\mathbf{V}_1 - [50j\ \Omega]\mathbf{I}_1 + [83.852j\ \Omega]\mathbf{I}_2 = 0$$

$$\mathbf{V}_1 + [83.852j\ \Omega]\mathbf{I}_2 = [50j\ \Omega]\mathbf{I}_1$$

Converting to polar form for easier multiplication:

$$[50j\ \Omega]\mathbf{I}_1 = (50\angle 90^\circ)(3\angle 45^\circ) = 150\angle 135^\circ = -106.07 + 106.07j$$

$$\implies \mathbf{V}_1 + (83.852j)\mathbf{I}_2 = (-106.07 + 106.07j)$$

KVL to mesh 2:

$$-\mathbf{I}_2 R - \mathbf{Z}_{L2}\mathbf{I}_2 + \mathbf{Z}_M\mathbf{I}_1 = 0$$

$$-\mathbf{I}_2[100\ \Omega] - [250j\ \Omega]\mathbf{I}_2 + [83.852j\ \Omega]\mathbf{I}_1 = 0$$

$$0\mathbf{V}_1 + (-100 - 250j)\mathbf{I}_2 = -(83.852j)\mathbf{I}_1$$

Converting to polar form for easier multiplication:

$$-(83.852j)\mathbf{I}_1 = -(83.852\angle 90^\circ)(3\angle 45^\circ) = -251.556\angle 135^\circ = 177.87 - 177.87j$$

$$\implies 0\mathbf{V}_1 + (-100 - 250j)\mathbf{I}_2 = (177.87 - 177.87j)$$

Solving this system of equations, we get:

$$\mathbf{V}_1 = -34.067 + 75.211j\ \text{V} \implies 82.56\angle 114.36^\circ$$

$$\mathbf{I}_2 = 0.368 + 0.858j\ \text{A} \implies 0.933\angle 66.78^\circ$$

$$\implies \mathbf{V}_2 = \mathbf{I}_2 R = 93.3\angle 66.78^\circ\ \text{V}$$

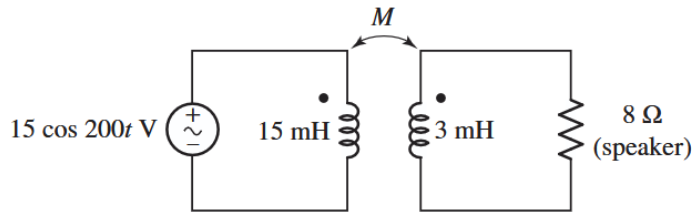
Mapping back to the original diagram, we have:

$$\boxed{\mathbf{V}_1 = 82.56\angle 114.36^\circ}$$

$$\boxed{\mathbf{I}_2 = -0.933\angle 66.78^\circ}$$

$$\boxed{\mathbf{V}_2 = -93.3\angle 66.78^\circ\ \text{V}}$$

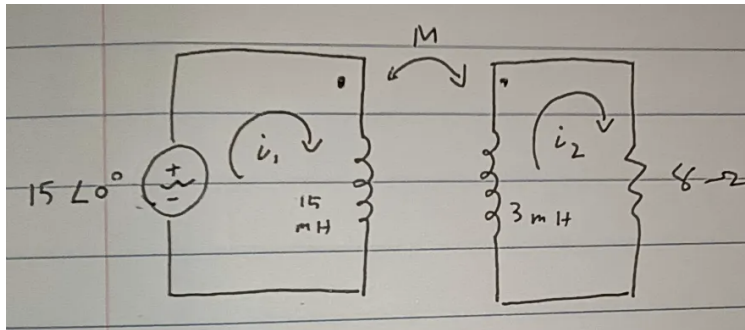
- 13.17** 17. The circuit of Fig. 13.45 is designed to drive a simple $8\ \Omega$ speaker. What value of M results in 1 W of average power being delivered to the speaker?



■ **FIGURE 13.45**

Defining impedances and constants:

- $\mathbf{V}_s = 15\angle 0^\circ\ \text{V}$
- $\omega = 200\ \text{rad/s}$
- $L_1 = 15\ \text{mH} \Rightarrow \mathbf{Z}_{L1} = j\omega L_1 = 3j\ \Omega$
- $L_2 = 3\ \text{mH} \Rightarrow \mathbf{Z}_{L2} = j\omega L_2 = 0.6j\ \Omega$
- $M \leq \sqrt{L_1 L_2} = 6.7\ \text{mH} \Rightarrow \mathbf{Z}_M = j\omega M = 200Mj\ \Omega$



The average power delivered to the speaker is given by:

$$P_{\text{avg}} = I_{\text{rms}}^2 R = \frac{I_{\text{max}}^2 R}{2}, \quad I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}}$$

$$\Rightarrow P_{\text{avg}} = \frac{1}{2} |\mathbf{I}_2|^2 R$$

$$\Rightarrow |\mathbf{I}_2|^2 = \frac{2P_{\text{avg}}}{R} = \frac{2[1\ \text{W}]}{[8\ \Omega]} \Rightarrow |\mathbf{I}_2| = 0.5$$

Thus, using mesh analysis we can solve for $\mathbf{I}_2$ and then calculate P_{avg} in terms of M :

KVL to mesh 1:

$$\mathbf{V}_s - \mathbf{Z}_{L1} \mathbf{I}_1 + \mathbf{Z}_M \mathbf{I}_2 = 0$$

$$(15) - (3j) \mathbf{I}_1 + (200Mj) \mathbf{I}_2 = 0$$

$$\mathbf{I}_1 = \frac{15}{3j} + \frac{200Mj}{3j}\mathbf{I}_2 = -5j + 66.66M\mathbf{I}_2$$

KVL to mesh 2:

$$-\mathbf{I}_2 R - \mathbf{Z}_{L2}\mathbf{I}_2 + \mathbf{Z}_M\mathbf{I}_1 = 0$$

$$-(8)\mathbf{I}_2 - (0.6j)\mathbf{I}_2 + (200Mj)\mathbf{I}_1 = 0$$

Plug in $\mathbf{I}_1$ from the first equation:

$$(-8 - 0.6j)\mathbf{I}_2 + (200Mj)(-5j + 66.66M\mathbf{I}_2) = 0$$

$$(-8 - 0.6j)\mathbf{I}_2 + 1000M + 13333M^2j\mathbf{I}_2 = 0$$

$$\mathbf{I}_2 = \frac{-1000M}{-8 - 0.6j + 13333M^2j}$$

Now find magnitude of $\mathbf{I}_2$:

$$|\mathbf{I}_2| = \frac{|-1000M|}{|-8 - 0.6j + 13333M^2j|} = \frac{1000M}{\sqrt{(-8)^2 + (-0.6 + 13333M^2)^2}} = 0.5$$

Solve for M :

$$\frac{10^6 M^2}{(-8)^2 + (-0.6 + 13333M^2)^2} = 0.25$$

$$10^6 M^2 = 0.25[(-8)^2 + (-0.6 + 13333M^2)^2]$$

$$10^6 M^2 = 0.25[64 + 0.36 - 1.59 \times 10^4 M^2 + 1.77 \times 10^9 M^4]$$

$$\implies 4.442 \times 10^8 M^4 - 1.00 \times 10^6 M^2 + 16 = 0$$

Using the quadratic formula to solve for $x = M^2$:

$$\implies 4.442 \times 10^8 x^2 - 1.00 \times 10^6 x + 16 = 0$$

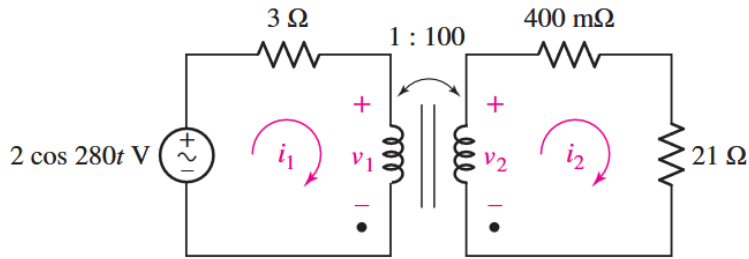
$$x = \frac{10^6 \pm \sqrt{10^{12} - 4(4.442 \times 10^8)(16)}}{2(4.442 \times 10^8)}$$

$$\implies M^2 = x = 2.235 \times 10^{-3}, 0.016 \times 10^{-3}$$

Since M^2 cannot be greater than $L_1 L_2 = 0.045 \times 10^{-3}$, we have:

$$M^2 = 0.016 \times 10^{-3} \implies \boxed{M = 4.014 \text{ mH}}$$

- 13.41** 41. Calculate the average power delivered to the $400\text{ m}\Omega$ and $21\text{ }\Omega$ resistors, respectively, in the circuit of Fig. 13.62.



■ **FIGURE 13.62**

Here, we can use the turns ratio a to find the reflected impedance $\mathbf{Z}_{\text{in}}$:

$$a^2 = \frac{\mathbf{Z}_{\text{L}}}{\mathbf{Z}_{\text{in}}}$$

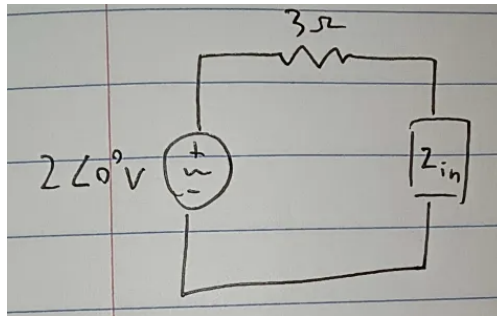
Where $a = 100$ and

$$\mathbf{Z}_{\text{L}} = [0.4\text{ }\Omega] + [21\text{ }\Omega] = [21.4\text{ }\Omega]$$

Thus,

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{Z}_{\text{L}}}{a^2} = \frac{21.4}{100^2} = 0.00214\text{ }\Omega$$

We can now redraw the circuit with the reflected impedance to find the current $\mathbf{I}_1$:



$$\Rightarrow \mathbf{Z}_{\text{eq}} = [3\text{ }\Omega] + \mathbf{Z}_{\text{in}} = [3.00214\text{ }\Omega]$$

$$\mathbf{I}_1 = \frac{\mathbf{V}_s}{\mathbf{Z}_{\text{eq}}} = \frac{2\angle 0^\circ}{3.00214} = 0.666\angle 0^\circ\text{ A}$$

Using turns ratio, we can find $\mathbf{I}_2$:

$$\frac{1}{a} = \frac{\mathbf{I}_2}{\mathbf{I}_1}$$

$$\Rightarrow \mathbf{I}_2 = \frac{\mathbf{I}_1}{a} = \frac{0.666\angle 0^\circ}{100} = 6.66 \times 10^{-3}\angle 0^\circ\text{ A}$$

Finally, we can directly calculate the average power dissipated by each resistor:

$$P_{\text{avg}} = \frac{1}{2} |\mathbf{I}|^2 R$$

- $P_{\text{avg}, 3\Omega} = \frac{1}{2}[0.666 \text{ A}]^2[3 \Omega] = \boxed{0.665 \text{ W}}$
- $P_{\text{avg}, 0.4\Omega} = \frac{1}{2}[6.66 \times 10^{-3} \text{ A}]^2[0.4 \Omega] = \boxed{8.87 \times 10^{-6} \text{ W}}$
- $P_{\text{avg}, 21\Omega} = \frac{1}{2}[6.66 \times 10^{-3} \text{ A}]^2[21 \Omega] = \boxed{4.65 \times 10^{-4} \text{ W}}$